Alexandre Caron

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EDUCATION

B.Sc. Honours Mathematics and Physics Fall 2024 – Winter 2027 (expected)
McGill University
CGPA 3.97/4.00. Dean's Honour List, 2025–2026.

ACADEMIC EXPERIENCE

Undergraduate Bachelor Thesis Fall 2026 – present
McGill University, Department of Physics. Supervisor: Prof. Simon Caron-Huot
Study of the duality between nearly-AdS_2 spacetime, which describes a simple traversable wormhole, and the coupled Sachdev–Ye–Kitaev (SYK) model. The project reviews existing numerical simulation methods for the coupled system and aims to improve them, in order to better predict when and how emergent gravitational behavior appears and to help guide future experiments.

Undergraduate Summer Research Project May 2026 – August 2026
McGill University, Department of Physics. Supervisor: Prof. Keshav Dasgupta
Studied the Unruh and Hawking effects from two perspectives: quantum field theory in curved spacetime, and the approach of Parikh and Wilczek, which treats black hole radiation as a tunneling process. Building on the second formalism, independently formulated a possible original derivation of the Unruh effect and investigated its physical consequences. Wrote a paper on the subject, submitted to the McGill Science Undergraduate Research Journal (MSURJ)

Directed Reading Program January 2026 – April 2026
McGill University, Department of Mathematics and Statistics. Supervisor: Shereen Elaidi (Ph.D. student); in collaboration with Xavier Zurowski
Studied how Lie groups and Lie algebras are used to solve differential equations, including exact solution methods for large classes of equations and the theory of solvability of ordinary differential equations. Co-wrote a report introducing the subject and presented the material in a talk to the Department.

Physics Tutor Winter 2024
Cégep de Saint-Hyacinthe
Held three one-on-one tutoring sessions per week with students, reviewing course concepts and working through problems.

PUBLICATIONS AND REPORTS

Submitted manuscript

- Caron, A. (2026). *On the Radiation of Horizons: Two Perspectives*. Manuscript submitted for publication.

Unpublished report

- Caron, A., & Zurowski, X. (2026). *Differential Equations and Symmetries*. Department of Mathematics and Statistics, McGill University.

TALKS

Caron, A., & Zurowski, X. (2026, April 30). *Symmetries and Differential Equations* [Oral presentation]. Department of Mathematics and Statistics, McGill University, Montréal, Québec, Canada.

AWARDS

- **Dean's Honour List**, McGill University, 2025–2026. Awarded to students in the top 10% of their faculty for the academic year.
- **Laureate, AMQ Math Competition**, Winter 2024. Placed 24th in Quebec in the annual 3-hour competition held by the Association Mathématique du Québec in cégeps across the province.

SKILLS AND INTERESTS

Programming. Python (NumPy, Matplotlib, Pandas, SciPy). C++: simple 2D simulations using SFML, including n gravitationally-interacting bodies with collisions and systems of thousands of non-interacting particles in a central potential. Also solved problems on Project Euler, a library of fun programming challenges.

Music. Piano for about 10 years; I taught myself composition, and like writing original pieces with MuseScore.